

Assignment – 2

Unit -1

1. Let X be a continuous random variable with p.d.f $f_X(x) = kx(1-x)$, $0 \leq x \leq 1$. Find k and determine a number b such that $P(X \leq b) = P(X \geq b)$. also obtain distribution function.

Ans: $k=6, b=1/2$, $F_X(x) = 3x^2 - 2x^3$

2. A random variable X has the following distribution function. Find the probability density function.

$$F_X(x) = 1 - e^{-x^2/b}$$

$$b > 0, x > 0.$$

Ans: $(2x/b)(e^{-x^2/b})$

3. The power reflected by an aircraft that is received by a radar can be described by an exponential random variable X. the probability density of X, is given by

$$f_X(x) = \begin{cases} \frac{1}{x_0} e^{-x/x_0}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Where x_0 is the average power received by the radar.

- (i) What is the probability that the radar will receive power larger than the power received on the average? Ans: 0.37
(ii) What is the probability that the radar will receive power less than the power received on the average? Ans: 0.63

4. The probability that a person will die in the time interval (t_1, t_2) is given by

$$P(t_1 \leq t \leq t_2) = \int_{t_1}^{t_2} \alpha(t) dt$$

$$\text{Where } \alpha(t) \begin{cases} = 3 \times 10^{-9} t^2 (100-t)^2, & 0 \leq t \leq 100 \\ 0, & \text{elsewhere} \end{cases}$$

Find (i) the probability that a person will die between the ages 60 years and 70 years. Ans: 0.154 (ii) the probability that he will die between the ages 60 and 70 years assuming that he lived upto 60 years. Ans: 0.486

5. Suppose that in a certain region the daily rainfall (In inches) is a continuous random variable X with probability density function $f_X(x)$, given by

$$f_X(x) \begin{cases} = \frac{3}{4}(2x-x^2), & 0 < x < 2 \\ 0, & \text{elsewhere} \end{cases}$$

Find the probability that on a given day in this region, the rainfall is:

- (i) Not more than 1 inch. Ans: $1/2$
- (ii) Greater than 1.5 inch. Ans: $5/32$
- (iii) Between 0.5 and 1.5 inch. Ans: $11/16$

6. If $f_X(x) = k(1+x)$, $2 \leq x \leq 5$, find $P(x < 4)$. Ans : $16/27$

7. Find the normalising factor, k in the following function is a probability density function.

$$f_X(x) = \begin{cases} k(1-x^2), & 0 < x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Also find $P(0.1 < x < 0.2)$ and $P(x > 0.5)$. ans: $k=3/2$, 0.146, 0.313